

ReinforcedLN: a Reinforcement Learning Approach for a Light Novel Recommender System

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Contents

1 Introduction 3
2 Market Analysis and Potential Impact 3
3 Existing Competition & Limitations 3
4 Proposed Solution 3
5 Technical Approach 4
6 Conclusion 4
References 5

1. Introduction

Modern recommender systems help users navigate vast digital libraries on platforms like Kindle, MyAnimeList, YouTube, Netflix, and Amazon. Yet they often rely on explicit user data—such as search history or purchase records—and require a known reference title. For light novels, solutions like Anibrain.ai(AniBrain 2025) depend on an initial title, limiting users who remember only general themes. Moreover, traditional search engines struggle with detailed narrative queries beyond brief keywords. The challenge is to create a system that accepts natural language descriptions of story ideas and returns relevant light novel suggestions without requiring prior title knowledge.

2. Market Analysis and Potential Impact

The light novel market is rapidly expanding, especially in Japan, with over 3,000 new titles published annually (Novels 2016). These illustrated narratives are popular among young adults and international readers. Although platforms like MyAnimeList and Novel Updates exist, most systems depend on known titles or simple keyword filters that miss the nuance of creative descriptions. Our system, which accepts free-form synopses, offers a more intuitive content discovery method, guiding new readers, uncovering niche titles, and delivering valuable market insights.

3. Existing Competition & Limitations

Current tools have notable shortcomings. Anibrain.ai, for example, only suggests titles similar to a provided reference, excluding users without one. Generic search engines handle short, generic queries well but falter with detailed descriptions. Likewise, tag-based filtering on platforms like MyAnimeList requires manual selection and fails to capture complex narrative details. Our approach overcomes these limitations by combining reinforcement learning with large language models to interpret detailed story descriptions and provide highly personalized recommendations.

4. Proposed Solution

Our system leverages reinforcement learning and advanced natural language processing to deliver dynamic, personalized light novel recommendations. It employs SBERT to encode user-entered narrative descriptions and the content's genres and synopses into dense embeddings, allowing for semantic matching that transcends simple keyword correlations. An epsilon-greedy policy facilitates balanced exploration, ensuring the system can discover and suggest new or unexpected titles, while a diversity penalty prevents repetitive outputs. This design not only captures the nuance in free-text

inputs but also continuously adapts based on user feedback, refining its recommendation strategy over time to deliver consistently relevant and varied suggestions.

5. Technical Approach

Data is sourced from the MyAnimeList API (MAL 2017), generating a dataset of over 10,000 light novels with attributes such as ID, title, cover images, mean score, rank, popularity, genres, and synopsis. For the Minimum Viable Product, this dataset is streamlined to around 2,000 entries, maintaining an 80:20 ratio of novels with and without synopses and balanced genre representation. The textual data is converted into dense embeddings using SBERT. A reinforcement learning agent, optimised with Proximal Policy Optimisation, concatenates the user's free-text embedding with each candidate's embedding, scoring and ranking them accordingly. A Gradio-based interface enables users to input story ideas, view the top three recommendations with details, and provide feedback, thereby continuously improving the model.

6. Conclusion

This innovative recommender system overcomes traditional content and keyword search limitations by utilising free-text input and dynamic user feedback. Beyond light novels, it could enhance personalisation on platforms like Netflix, YouTube, Amazon, and Spotify.

References

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